

Vehicle Control Engineering (70150113)

Final Project

1. Background

With the rapid advancement of autonomous driving, the core technologies of decision-making, planning, and motion control for intelligent vehicles have become paramount. This project is designed to allow students to apply vehicle control theory in a high-fidelity dynamic simulator. You will design, implement, and validate your own algorithm capable of navigating a challenging racing scenario, pushing the boundaries of both safety and driving performance.

2. Project Description

2.1 Core Objective

The primary objective is to design and implement a robust **decision, planning and control algorithm** (or Integrated Decision and Control, IDC) that guides an autonomous vehicle (Ego Vehicle) to complete **one full lap** on a fixed, closed-loop racing map.

This task is a comprehensive challenge: the autonomous vehicle must operate **safely** (avoiding all obstacles), **compliantly** (obeying traffic laws), and **quickly** (minimizing total lap time).

2.2 The Scenario

You will be operating in a dynamic urban racing environment with the following characteristics:

- **Closed-Loop Map:** The track is a continuous "ring road." Your goal is to start, complete one full circuit, and finish.
- **Dynamic Obstacles:** The map includes other traffic participants. These vehicles follow fixed, predefined trajectories and **will not** react to your Ego Vehicle. You are solely responsible for anticipating their movements and navigating around them.
- **Road Infrastructure:** The map includes traffic infrastructure, most importantly **traffic lights**, which must be obeyed.

Your algorithm must perceive this environment, make real-time decisions, and execute smooth, precise control to balance all competing objectives.

Metric	Description	Measurement and Scoring (Lower is better, except for Pass/Fail)
1. Safety Score (Prerequisite)	Pass/Fail. Checks for collisions and off-road events.	1.0 (Pass): No safety violations. 0.0 (Fail): Any collision or off-road event occurred.
2. Compliance Score (Prerequisite)	Pass/Fail. Checks for traffic light violations.	1.0 (Pass): No traffic light violations. 0.0 (Fail): Any red light was run.
3. Racing Score (Primary Perf. Metric)	Total lap time (in seconds). This is the primary ranking metric.	Calculated as <code>total_steps * dt</code> . Lower is better.
4. Comfort Score (Secondary Perf. Metric)	Measures ride smoothness (jerk, acceleration).	Calculated from the mean of action values and their differences (e.g., <code>acc_diff_mean</code> , <code>steer_abs_mean</code>). Lower is better.
5. Algorithm Efficiency (Secondary Perf. Metric)	Measures computational performance.	The average execution time (in milliseconds) of your <code>agent.update</code> function. Lower is better.

6. Submission Contents

Please package the following into a single `.zip` file (one submission per team):

- Code:** All `.py` files you modified. Please use git to manage the project. Each student's code contribution will be reflected in the git commit.
- Video (`video.mp4`):** A screen recording of your best lap and the simulation address.
- Report (`report.pdf`):** A final report in IEEE Journal template. Project **report** is no more than 10 pages (strictly required).
- PPT:** Final presentation PPT (strictly follow the template).

7. Technical Documentation

All detailed technical information, including environment setup, code interfaces, and API definitions, is provided in separate documents.

- `Installation_Guide`: Refer to this for complete environment setup and verification steps.
- `Technical_Guide`: Refer to this for detailed explanations of:
 - The LasVSim platform-agent interface.
 - The structure of the observation dictionary.
 - Definitions of all metrics and helper functions.

